

PASSIVE KEYLESS ENTRY

High-Level Design Document

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Status: Approved for Engineering Analysis

Classification: Synthetic Demonstration Data

1. System Overview

The Passive Keyless Entry coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are AccessManager, KeyFobControl, DoorLockControl, ImmobilizerControl, and ProximitySensorManager.

3. Software Components

Component	Role	Primary Responsibility
AccessManager	Coordinator	Authorises entry and engine start.
KeyFobControl	Provider	Receives and decodes key fob messages.
DoorLockControl	Consumer	Locks and unlocks the doors.
ImmobilizerControl	Consumer	Releases engine start authorisation.
ProximitySensorManager	Provider	Estimates the key distance.

4. Interfaces

Interface	Provider	Consumer	Signals
IKeyMessage	KeyFobControl	AccessManager	KeyChallenge
IKeyDistance	ProximitySensorManager	AccessManager	KeyDistance
ILockRequest	AccessManager	DoorLockControl	LockState
IStartRelease	AccessManager	ImmobilizerControl	StartAuthorisation

5. Ports

Component	Port	Direction	Interface
KeyFobControl	KeyMessage_Out	P-PORT	IKeyMessage
AccessManager	KeyMessage_In	R-PORT	IKeyMessage
ProximitySensorManager	KeyDistance_Out	P-PORT	IKeyDistance
AccessManager	KeyDistance_In	R-PORT	IKeyDistance
AccessManager	LockRequest_Out	P-PORT	ILockRequest
DoorLockControl	LockRequest_In	R-PORT	ILockRequest
AccessManager	StartRelease_Out	P-PORT	IStartRelease
ImmobilizerControl	StartRelease_In	R-PORT	IStartRelease

6. Signals

Signal	Data Type	Source	Destination
KeyChallenge	uint32	KeyFobControl	AccessManager
KeyDistance	uint8	ProximitySensorManager	AccessManager
LockState	boolean	AccessManager	DoorLockControl
StartAuthorisation	boolean	AccessManager	ImmobilizerControl

7. Dependencies

Source	Relationship	Target	Reason
KeyFobControl	PROVIDES_INTERFACE	IKeyMessage	Publishes key challenge.
AccessManager	REQUIRES_INTERFACE	IKeyMessage	Consumes key challenge.
ProximitySensorManager	PROVIDES_INTERFACE	IKeyDistance	Publishes key distance.
AccessManager	REQUIRES_INTERFACE	IKeyDistance	Consumes key distance.
AccessManager	PROVIDES_INTERFACE	ILockRequest	Publishes lock state.
DoorLockControl	REQUIRES_INTERFACE	ILockRequest	Consumes lock state.
AccessManager	PROVIDES_INTERFACE	IStartRelease	Publishes start authorisation.
ImmobilizerControl	REQUIRES_INTERFACE	IStartRelease	Consumes start authorisation.

8. Functional Flows

Flow F-001: Key Challenge Distribution

KeyFobControl publishes KeyChallenge through IKeyMessage. AccessManager receives the signal through KeyMessage_In and updates its internal state model.

Flow F-002: Key Distance Distribution

ProximitySensorManager publishes KeyDistance through IKeyDistance. AccessManager receives the signal through KeyDistance_In and updates its internal state model.

Flow F-003: Lock State Distribution

AccessManager publishes LockState through ILockRequest. DoorLockControl receives the signal through LockRequest_In and updates its internal state model.

Flow F-004: Start Authorisation Distribution

AccessManager publishes StartAuthorisation through IStartRelease. ImmobilizerControl receives the signal through StartRelease_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.